

ERNAGE, Belgium

WMO: 064590

Lat: 50.5819N

Lon: 4.6892E

Elev: 157

StdP: 99.45

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-5.0	-9.9	1.6	-7.0	-6.3	2.3	-4.0	13.1	8.6	12.1	7.3	3.1	60	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.2	20.7	26.8	19.7	24.9	18.7	22.0	27.0	20.7	25.0	19.6	23.5	3.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	15.2	24.0	19.2	14.3	22.8	18.1	13.3	21.4	65.0	27.4	60.3	24.9	56.4	23.4	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.4	(b) 9.3	(c) 8.2	(e) -10.5	(f) 33.4	(g) 5.3	(h) 2.5	(i) -14.4	(j) 35.3	(k) -17.5	(l) 36.7	(m) -20.5	(n) 38.1	(o) -24.4	(p) 40.0		
(a) 10.4	(b) 9.3	(c) 8.2	(e) -10.7	(f) 23.9	(g) 5.4	(h) 1.6	(i) -14.5	(j) 25.0	(k) -17.7	(l) 25.9	(m) -20.7	(n) 26.8	(o) -24.6	(p) 28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	2.7	3.2	6.4	9.5	12.9	15.9	18.0	17.7	14.5	10.9	6.8	4.1
	DBStd	6.43	4.32	4.17	3.63	3.57	3.64	3.13	3.23	2.76	2.79	3.52	3.52	3.99
	HDD10.0	926	226	192	119	52	12	0	0	0	1	31	107	186
	HDD18.3	3029	483	425	369	266	171	87	45	45	119	230	346	442
	CDD10.0	1021	1	0	8	36	103	178	249	237	136	59	11	2
	CDD18.3	83	0	0	0	0	4	14	36	24	5	1	0	0
	CDH23.3	816	0	0	0	5	38	139	344	242	45	2	0	0
	CDH26.7	224	0	0	0	0	5	31	103	76	9	0	0	0
Wind	WSAvg	4.0	4.7	4.6	4.6	3.8	3.7	3.6	3.5	3.5	3.5	3.8	4.3	4.8
Precipitation	PrecAvg	808	63	53	68	56	70	79	69	69	61	69	76	74
	PrecMax	1098	148	109	140	131	140	160	169	145	166	160	126	186
	PrecMin	454	22	7	18	13	19	9	22	17	14	6	23	13
	PrecStd	147	30	28	33	31	33	39	37	34	36	43	25	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.5	18.9	23.9	26.7	30.4	33.0	32.6	27.9	22.9	16.9	13.2
		MCWB	11.4	8.4	12.7	14.8	19.1	21.3	21.1	21.0	20.3	16.6	15.2	11.9
	2%	DB	11.0	10.9	16.5	20.9	24.0	26.9	29.3	28.7	24.1	19.8	14.9	11.9
		MCWB	10.3	9.2	11.6	14.1	17.0	20.0	21.2	20.5	18.1	16.7	13.7	10.7
	5%	DB	9.5	9.8	13.9	18.1	21.9	24.6	26.8	25.9	21.7	17.6	13.5	10.7
		MCWB	8.8	8.6	10.8	13.2	16.5	18.9	19.5	19.4	17.3	15.2	12.5	10.0
	10%	DB	8.3	8.7	11.8	15.7	19.8	22.4	24.8	23.5	19.6	16.0	12.1	9.4
		MCWB	7.7	7.7	9.3	11.8	15.7	17.9	18.9	18.4	15.9	14.5	11.2	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	10.5	13.5	16.4	20.6	22.8	23.6	23.6	20.9	18.4	15.7	12.4
		MCDB	12.0	11.9	17.0	21.7	24.6	28.2	29.0	29.6	26.7	21.3	16.2	12.9
	2%	WB	10.4	9.5	12.2	14.7	18.5	20.9	22.2	21.4	19.0	16.9	13.8	11.0
		MCDB	10.9	10.4	15.8	19.0	21.9	24.8	27.6	25.9	22.9	18.8	14.6	11.6
	5%	WB	8.9	8.7	11.0	13.4	17.1	19.5	20.9	20.0	17.8	15.7	12.6	10.0
		MCDB	9.4	9.6	13.4	17.1	20.4	23.5	25.5	23.8	20.4	17.2	13.5	10.7
	10%	WB	7.7	7.7	9.8	12.2	16.0	18.3	19.6	18.9	16.6	14.6	11.3	8.8
		MCDB	8.2	8.7	11.5	15.5	19.3	21.9	23.6	22.8	18.9	15.9	12.0	9.4
Mean Daily Temperature Range	5% DB	MDBR	4.3	5.1	7.3	9.2	9.4	9.8	10.0	9.6	8.6	7.0	5.0	4.2
		MCDBR	4.8	7.1	11.4	12.8	12.6	13.4	13.9	14.2	11.9	9.2	6.6	5.0
		MCWBR	4.4	5.4	7.4	7.3	6.5	7.3	6.6	6.9	6.5	6.1	5.5	4.3
	5% WB	MCDBR	4.8	6.3	10.0	11.3	10.6	11.8	12.3	11.6	10.1	8.3	6.1	4.9
		MCWBR	4.4	5.1	6.9	6.8	6.4	7.1	6.7	6.7	6.3	5.8	5.2	4.5
Clear-Sky Solar Irradiance	taub	0.319	0.341	0.374	0.409	0.419	0.416	0.419	0.411	0.382	0.357	0.332	0.316	
	taud	2.412	2.375	2.286	2.205	2.218	2.245	2.239	2.285	2.353	2.419	2.414	2.381	
	Ebn at Noon	712	787	821	830	838	842	833	820	805	757	684	648	
	Edh at Noon	62	83	107	130	134	132	131	119	100	78	62	56	
All-Sky Solar Radiation	RadAvg	0.78	1.45	2.60	4.13	4.90	5.34	5.08	4.32	3.27	1.93	0.92	0.62	
	RadStd	0.11	0.29	0.37	0.48	0.49	0.48	0.61	0.46	0.34	0.16	0.11	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page